

Mini-project 5.3: Detecting the Anomalous Activity of a Ship's Engine

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1 Introduction

We have analyzed a dataset monitoring critical ship engine parameters to detect operational anomalies. Engine malfunctions present serious risks, including potential safety hazards for crew members and operational downtime that can disrupt shipping schedules, and therefore business performance. Implementing predictive maintenance through anomaly detection aims to optimize operations by reducing inefficiencies and safety risks, while ensuring reliable performance.

2 Objectives

The task is to develop a robust anomaly detection system to protect a company's shipping fleet by evaluating engines functionality.

- Explore the data and employ preprocessing and feature engineering (if required).
- Using statistical techniques and machine learning algorithms, perform anomaly detection, post-processing, and refinement.
- Write a report with critical findings and recommendations.

3 Methods

A dataset containing six real-time measurements of ship engine performance (engine RPM, lubrication oil pressure, fuel pressure, coolant pressure, lubrication oil temperature, and coolant temperature) was obtained from this github repository.

The data was preprocessed to address missing or invalid values (duplicates or formatting errors). As the analysis focused on the six provided features, no feature engineering was performed. Exploratory data analysis (EDA) confirmed the dataset was in good condition for further analysis.

The analysis used Python 3.12.0 with standard libraries (pandas, NumPy, Matplotlib, Seaborn) and scikit-learn's OneClassSVM and IsolationForest for anomaly detection. Interquartile range (IQR) analysis was implemented using pandas' quantile method and custom code. Additional libraries were used for

visualization and data processing. The full code is available in the Python notebook accompanying this report and can also be found [here](#).

4 Results

Exploratory Data Analysis (EDA) revealed the dataset is well-structured and suitable for analysis. While all features contained duplicate values, duplication rates were below 1% for most variables. The exception was Engine RPM, with 1,206 duplicates out of 19,535 values (6.2%). These duplicates reflect actual engine operating ranges rather than data entry errors, and thus do not compromise analysis validity. Subsequent analysis proceeded under this assumption.

The summary statistics (Table 1) indicate that all features exhibit slight right skewness, with medians positioned left of their respective means, a pattern consistent with the boxplot distributions (Figure 1). Analysis of 95th percentile thresholds identified 678 multivariate outliers where at least two features simultaneously exceeded extreme value thresholds. These joint extreme events are particularly valuable for understanding feature relationships under abnormal operating conditions.

In future analysis, it might be of interest to compute summary statistics for the extreme values to better understand their distribution and gain deeper insights into the behavior these of extreme values and their implications.

Table 1: Summary Statistics for the Dataset

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp
count	19535.00	19535.00	19535.00	19535.00	19535.00	19535.00
mean	791.24	3.30	6.66	2.34	77.64	78.43
std	267.61	1.02	2.76	1.04	3.11	6.21
min	61.00	0.00	0.00	0.00	71.32	61.67
25%	593.00	2.52	4.92	1.60	75.73	73.90
50%	746.00	3.16	6.20	2.17	76.82	78.35
75%	934.00	4.06	7.74	2.85	78.07	82.92
max	2239.00	7.27	21.14	7.48	89.58	195.53

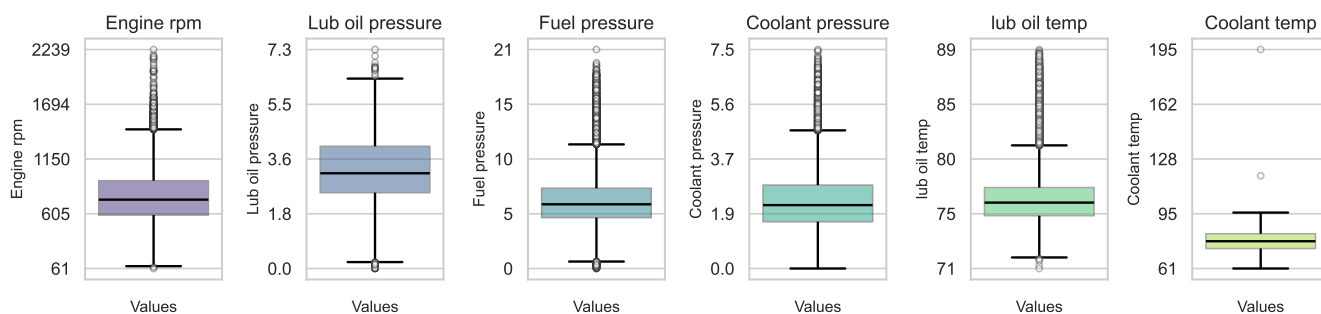


Figure 1: Distribution of engine parameters across all measurements: Box plots demonstrating the distribution of engine parameters across all measurements.

4.1 Anomaly Detection

In statistical terms, anomaly or outlier detection identifies observations that ‘significantly’ deviate from a data set’s expected pattern or distribution. To achieve our goal of developing a robust anomaly detection system for evaluating the engines functionality of a company’s fleet of ships, three different outlier detection approaches were used.

EDA confirmed non-normal distribution across all features (See notebook for details). Consequently, the data was analyzed using anomaly detection methods that are robust to non-normal distributions.

4.1.1 Using IQR method

The IQR analysis identified 23.73% of observations as outliers in ≥ 1 feature, exceeding the recommended 1-5% threshold for this assignment. This suggests that this threshold is either overly permissive or that the data set contains a high proportion of extreme values in the data.

Conversely, only 0.06% of observations were flagged as outliers in ≥ 3 features simultaneously, indicating this threshold is too restrictive for meaningful anomaly detection.

Interestingly, 2.16% of observations were flagged as outliers in ≥ 2 features simultaneously, which falls within the ideal 1-5% range recommended for this assignment, suggesting this as the most balanced threshold for multivariate anomalies in this dataset.

Importantly, while IQR is robust to non-normality, it is suboptimal for multivariate detection as it ignores feature correlations. Therefore, these observations must be considered with caution.

4.1.2 Using One-Class SVM method (OCSVM)

When configured with an appropriate kernel, such as the Radial Basis Function (RBF) kernel, OCSVM is an effective algorithm for multivariate anomaly detection, as it is capable of dealing with non-normal data and capturing nonlinear relationships (Scikit-learn developers, 2024).

In this data set, OCSVM predicted 585 anomalies ($\gamma=0.1$ and $\nu=0.03$), which represent 3.0% of the total observations. This falls within the suggested range of 1-5% anomalies for the assignment. The number of anomalies detected can be increased or decreased by tuning ν and γ , so careful validation of the results is required before drawing meaningful conclusion. At this point, the number of anomalies detected closely align with IQR, suggesting that the performance is reasonable. Further analysis is needed to evaluate the method's performance.

For visualization, the dataset was reduced to two principal components (PC1 and PC2) using PCA. Figure 2 shows normal observations forming a regular-shaped cloud (blue dots), while anomalies scatter across, with some concentrated at the edges. Notably, one extreme outlier lies far from the main cloud, corresponding to the maximum Coolant temperature (See notebook for details). Though not a multivariate anomaly, it was flagged by both IQR and OCSVM, most likely due to the extreme value. No other clear pattern was observed in the PCA space.

Of note, PC1 and PC2 explained only 40% of the cumulative variance in this data set (See notebook for details), indicating that with this PCA projection a substantial portion of the data's variability is not being displayed. While useful for visualization, such low variance retention limits its utility for analyses requiring high feature fidelity.

4.1.3 Using Isolation Forest method

Isolation Forest (IF) is an unsupervised anomaly detection algorithm that identifies anomalies by randomly partitioning data using decision trees (Scikit-learn developers, 2024). Applied to this dataset with $n_estimators=200$ and $contamination=0.03$, IF identified 587 anomalies (3.0% of observations), aligning with the recommended 1–5% range for this assignment. Increasing $n_estimators$ from 100 to 200 yielded slight improvement in consistency with results from OCSVM. No significant benefit was observed with further increases in this parameter. The distribution of anomalies in the PCA space (Figure 2) closely matched the patterns observed with the IQR and OCSVM methods, suggesting robustness across different methodologies.

The similarity in anomaly patterns and distributions across the three methods raises an intriguing question: are these methods identifying the same anomalies? Although IF and OCSVM detected an equivalent number of anomalies, only 64% overlapped (Figure 3). The discrepancy grew when compared against IQR, showcasing 32% and 43% similarity with OCSVM and IF, respectively. These variations can be attributed to the distinct methodologies employed; IF excels at detecting local anomalies (isolated points), while OCSVM is more effective at identifying global anomalies (dense regions). IQR, conversely, is less suited for multivariate anomaly detection due to its reliance on univariate statistics.

Accordingly, evaluating these anomalies against business context is essential. The discrepancy between the three methods requires careful validation of the results, since it is likely that some anomalies may represent edge cases rather than true outliers.

Feature importance analysis could help refine the IF model by prioritizing high-impact variables. Nonetheless, for actionable insights, domain experts should review a sample of flagged anomalies to assess their relevance.

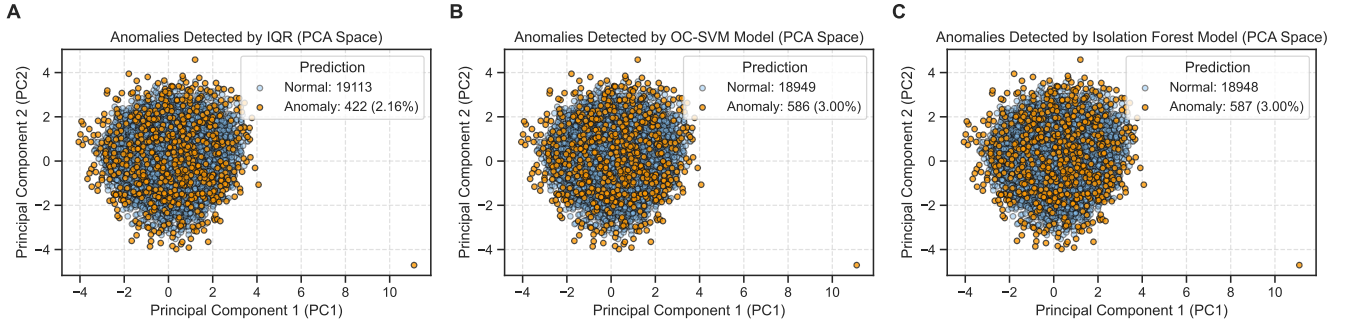


Figure 2: Visualization of Anomalies Detected by Three Methods: Scatter plots comparing normal and anomalous observations in PCA space, as identified by (A) the IQR method, (B) OCSVM, and (C) Isolation Forest (IF).

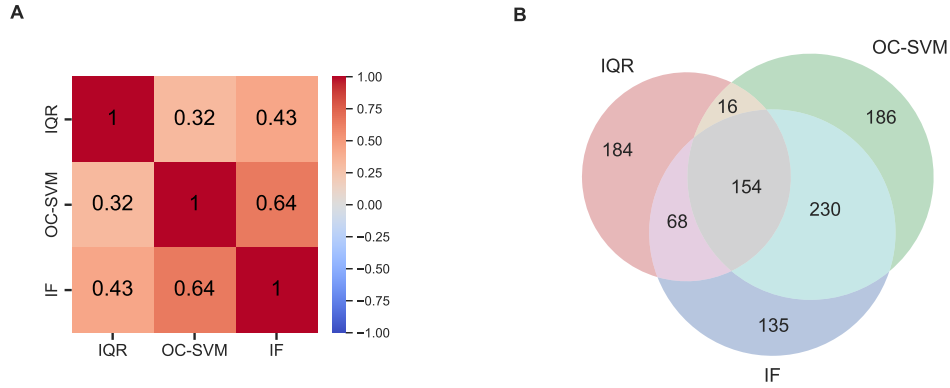


Figure 3: Comparison Anomalies detected by Three different approaches: (A) Correlation matrix showing pairwise agreement coefficients between the three anomaly detection methods: Interquartile Range (IQR), One-Class Support Vector Machine (OC-SVM), and Isolation Forest (IF). Values range from -1 (perfect inverse correlation) to +1 (perfect agreement), with 0 indicating no linear relationship. (B) Venn diagram quantifying the overlap in anomaly identifications between methods, revealing both consensus detections (intersections) and method-specific anomalies (unique sections).

5 Conclusions

Understanding the strengths and limitations of each method is crucial for selecting the optimal approach for specific anomaly detection tasks. Future work could investigate ensemble techniques to combine these methods and improve both detection accuracy and interpretability.

While IQR, OCSVM, and IF employ different principles, they all detect anomalies through feature-wise irregularities (e.g., extreme values or local patterns) rather than spatial relationships in 2D projections (Liu et al., 2008; Schölkopf et al., 2001). Although these methods operate differently in high-dimensional space,

their anomaly distributions may appear similarly scattered in PCA visualizations due to oversimplification of the data structure. Accordingly, the superficial similarity in PCA space observed in this analyses did not reflected identical anomaly detection, as demonstrated by their limited overlap (Figure 3).

Furthermore, since PC1 and PC2 capture only 40% of the variance, many anomalies likely result from patterns in higher principal components or non-linear relationships that PCA cannot represent. This highlights a fundamental challenge in multivariate anomaly detection, where observations may be anomalous in certain feature while appearing normal in others.

6 References

- Scikit-learn developers. (2024). Scikit-learn: Machine Learning in Python (Version 1.4.0) [Computer software]. <https://scikit-learn.org/stable/>
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